

TENG GAO

tgaoteng@gmail.com | [ORCID](#) | [GitHub](#) | [Google Scholar](#)

EDUCATION

- Harvard University** (Cambridge, MA, USA) Sep. 2020 – May 2024
Ph.D. in Bioinformatics and Integrative Genomics (GPA: 4.00/4.00)
Thesis advisors: Peter J. Park and Peter V. Kharchenko
Committee: Nir Hacohen, Vijay G. Sankaran, Kamila Naxerova
- Washington University** (St. Louis, MO, USA) Sep. 2014 – May 2018
B.S. Double Major in Computer Science and Biology (GPA: 3.99/4.00)
Summa Cum Laude
Honor thesis advisor: Christopher Maher

RESEARCH POSITIONS

- Boston Children's Hospital & Broad Institute of MIT and Harvard** Jun. 2024 – Present
Damon Runyon-HHMI Postdoctoral Fellow (Advisor: Vijay G. Sankaran)
• Studying hematopoietic stem cell dynamics in bone marrow regeneration using lineage-resolved single-cell multiomics
- Harvard Medical School, Department of Biomedical Informatics** Mar. 2021 – May 2024
Graduate Researcher (Advisors: Peter J. Park and Peter V. Kharchenko)
• Developed Numbat, a novel computational method to infer genome aberrations and single-cell phylogeny from scRNA-seq data
• Applied Numbat to characterize pediatric neuroblastoma and age-related mosaic chromosomal alterations in normal tissues
• Developing new computational methods for the analysis of subcellular RNA organization from spatial transcriptomics data
- Memorial Sloan Kettering Cancer Center, Computational Oncology** May 2018 – Aug. 2020
Computational Biologist I-II (Advisors: Elli Papaemmanuil and Kelly Bolton)
• Led large-scale study on clonal hematopoiesis and therapy-related leukemia in 32,000 cancer patients
- Washington University in St. Louis, McDonnell Genome Institute** Sep. 2016 – May 2018
Undergraduate Researcher (Advisor: Christopher Maher)
• Co-developed INTEGRATE-Vis, a Python visualization package for patient-specific gene fusions discovered in RNA-seq data
• Honor thesis: “ceFinder: Machine learning based prediction of novel competing endogenous RNAs.”
- Roche Molecular Systems** May 2017 – Aug. 2017
Bioinformatics Intern (Manager: Dwight Kuo)
• Developed RNA-seq analysis pipelines for novel miRNA biomarker discovery and differential expression in prostate cancer
• Identified a novel antibiotic resistance gene cassette in methicillin-resistant *S. aureus* through de novo genome assembly
- Wake Forest Institute for Regenerative Medicine** May 2016 – Aug. 2016
Summer Scholar (Advisors: Anthony Atala and Sang Jin Lee)
• Established a mathematical model for biomaterial viscoelasticity in 3D biofabrication

PUBLICATIONS

(*: equal contributions; ‡: co-corresponding authors)

Selected Publications

- Mitchel J*, Gao T*, Petukhov V*, Cole E, Kharchenko PV (2026). “Impact and correction of segmentation errors in spatial transcriptomics”. *Nature Genetics* 58, 434–444. 10.1038/s41588-025-02497-4. [[Paper](#)]
- Gao T, Eleni MK, Ljungström V, Heinzl A, Tischler AS, Oberbauer R, Loh PR, Adameyko I, Park PJ[‡], Kharchenko PV[‡] (2023). “A pan-tissue survey of mosaic chromosomal alterations in 948 individuals”. *Nature Genetics* 10.1038/s41588-023-01537-1. [[Paper](#)] [[Code](#)]

3. **Gao T**, Soldatov R, Sarkar H, Kurkiewicz A, Biederstedt E, Loh PR, and Kharchenko PV (2022). “Haplotype-aware analysis of somatic copy number variations from single-cell transcriptomes”. *Nature Biotechnology* 41, 417–426. 10.1038/s41587-022-01468-y. [\[Paper\]](#) [\[Code\]](#)
4. **Gao T**, Ptashkin R, Bolton KL, Sirenko M, Fong C, Spitzer B, Menghrajani K, Ossa JEA, Zhou Y, Bernard E, Levine M, Martinez JSM, Zhang Y, Franch SE, Patel M, Braunstein LZ, Kelly D, Yabe M, Benayed R, Caltabellotta NM, Philip J, Paraiso E, Mantha S, Solit DB, Diaz LA Jr, Berger MF, Klimek V, Levine RL, Zehir A, Devlin SM, Papaemmanuil E (2021). “Interplay between chromosomal alterations and gene mutations shapes the evolutionary trajectory of clonal hematopoiesis”. *Nature Communications* 12, 338. 10.1038/s41467-020-20565-7. [\[Paper\]](#) [\[Code\]](#)

Preprints

1. **Gao T**^{*‡}, Weng C^{*‡}, Johnson I, Poeschla M, Gudera J, King E, Rouya C, Donovan A, Bourke L, Shao Y, Marquez E, Tyagi R, Zon LI, Weissman JS[‡], Sankaran VG[‡] (2026). “Modeling mitochondrial inheritance enables high-precision single-cell lineage tracing in humans”. *bioRxiv* (preprint). 10.64898/2026.02.12.705660. [\[Paper\]](#) [\[Code\]](#)

Other Publications

1. Batchi-Bouyou AL, Tran D, Raddatz M, **Gao T**, Vlasschaert C, Sharber B, Liu J, Petrone G, Chan IC, Beeler JS, Mustion G, Bick AG, Devlin SM, Bolton KL (2026). “Predictive Modeling of Clonal Hematopoiesis Across Diverse Cohorts”. *Blood Advances* 10.1182/bloodadvances.2025017940. [\[Paper\]](#)
2. Dela Cruz FS, You D, Feinberg TY, Brosius S, Wang X, McCarter JG, Domenico D, Gutiérrez-Abril J, Arango Ossa JE, Gundem G, Walch HS, Chatila WK, Xu J, Li H, Guilan K, Siddiquee A, Silber J, Benhamida J, Linkov I, Ntiamoah P, Sauter JL, Bhanot UK, Jain M, Kombak FE, Medina-Martínez JS, Levine MF, Glodzik D, **Gao T**, Diolaiti D, F Coutinho D, Rose R, Li S, Stockfisch E, Bouvier N, Roberts RD, Yustein J, Rainusso N, Crompton BD, Ortiz MV, Slotkin EK, Thomas AD, Sait SF, Mattar MS, Meneses M, Rosales N, Kinnaman MD, Rodríguez-Sánchez MI, LaQuaglia MP, Gerstle JT, Iacobuzio-Donahue CA, Glade Bender JL, Roehrl MH, Schultz N, Mauguén A, de Stanchina E, Shukla NN, Papaemmanuil E, Kung AL (2026). “A Clinically Integrated Pediatric Patient-Derived Xenograft Program Enables Evaluation of Cohort and Patient-Specific Biology and Therapeutic Strategies”. *Cancer Research* 10.1158/0008-5472.CAN-25-3930. [\[Paper\]](#)
3. Poverennaya I, Murtazina A, Li L, Maili L, Sourada L, Montano-Gutierrez LF, Galimullina R, Steinschaden T, Kaiser M, Zikmund T, Goralija A, **Gao T**, Attina A, Clara O, Bartenhagen C, Erickson A, Gershtein Y, Chen S, Polaskova K, Sterba J, Semasch B, Anderson ER, Prakash V, Vincent T, Arceo M, Kogner P, Schlisio S, Kharchenko PV, David A, Kaiser J, Fischer M, Skoda J, Trainor PA, Chagin AS, Adameyko I (2026). “Ribosomal modifications are associated with mesenchymal fate selection in the neural crest lineage”. *Nature Communications* 17, 2326. 10.1038/s41467-026-70375-6. [\[Paper\]](#)
4. Della Volpe L, Lee AJ, Antoszewski M, Deik AA, Safina KR, **Gao T**, Guo CJ, Ye T, Lyu P, Martin-Rufino JD, Castano N, Good J, Molina-Aponte Y, Zhao J, Clish CB, van Galen P, Sankaran VG (2025). “Inhibiting ferroptosis enhances ex vivo expansion of human haematopoietic stem cells”. *Nature Cell Biology* 27, 2214–2224. 10.1038/s41556-025-01814-7. [\[Paper\]](#)
5. Zhang H, Zhang Y, **Gao T**, Wu Y (2025). “ScisTree2 enables large-scale inference of cell lineage trees and genotype calling using efficient local search”. *Genome Research* 35, 2781–2791. 10.1101/gr.280542.125. [\[Paper\]](#)
6. Li R, Alberge J-B, Keshavarzian T, Tsuji J, Gustafsson J, Rahmat M, Lightbody ED, Deng SL, Riviero S, Miller M, Kalayci FNC, Wiestner A, Sun C, Lupien M, Ghobrial I, Parry E, **Gao T**, Getz G (2025). “Numbat-multiome: inferring copy number variations by combining RNA and chromatin accessibility information from single-cell data”. *Briefings in Bioinformatics* 26, bbaf516. 10.1093/bib/bbaf516. [\[Paper\]](#)
7. Liu J, Tran D, Xue L, Wiley BJ, Vlasschaert C, Watson CJ, MacGregor HAJ, Zong X, Chan ICC, Das I, Uddin MM, Niroula A, Griffin G, Ebert BL, Mack T, Pershad Y, Sharber B, Berger M, Zehir A, Ptashkin R, Levine RL, Papaemmanuil E, Joseph V, **Gao T**, Kemel Y, Mandelker D, Stopsack KH, Pharoah PDP, Mukherjee S, Ding L, Cao Y, Walter MJ, Blundell JR, Chatterjee N, Offit K, Godley LA, Link DC, Stadler ZK, Bick AG, Natarajan P, Bolton KL (2025). “Germline genetic variation impacts clonal hematopoiesis landscape and progression to malignancy”. *Nature Genetics* 57, 1872–1880. 10.1038/s41588-025-02250-x. [\[Paper\]](#)
8. **Gao T**[‡], Sankaran VG[‡] (2024). “Disease Background Influences Fate of Transplanted Stem Cells.” *Nature* (News & Views), October 23. 10.1038/d41586-024-03112-y. [\[Paper\]](#)

9. Olsen TK, Otte J, Mei S, Embaie BT, Kameneva P, Cheng H, **Gao T**, Zachariadis V, Tsea I, Björklund Å, Kryukov E, Hou Z, Johansson A, Sundström E, Martinsson T, Fransson S, Stenman J, Fard SS, Johnsen JI, Kogner P, Adameyko I, Enge M, Kharchenko PV, Baryawno N (2024). “Joint single-cell genetic and transcriptomic analysis reveal pre-malignant SCP-like subclones in human neuroblastoma”. *Molecular Cancer* 23, 180 (2024). 10.1186/s12943-024-02091-y. [\[Paper\]](#)
10. Franch SE, Mehine M, Ptashkin RN, Bolton KL, Bandlamudi C, Srinivasan P, Zhang L, Goodell MA, Gedvilaite E, Menghrajani K, Sánchez-Vela P, Mandelker D, Comen E, Norton L, Benayed R, **Gao T**, Papaemmanuil E, Taylor B, Levine R, Offit K, Stadler Z, Berger MF, Zehir A (2023). “Associations between cancer predisposition mutations and clonal hematopoiesis in patients with solid tumors”. *JCO Precision Oncology* 2023:7. 10.1200/PO.23.00070. [\[Paper\]](#)
11. Spitzer B, Rutherford KD, Gundem G, McGovern EM, Millard NE, Arango Ossa JE, Cheung IY, **Gao T**, Levine MF, Zhang Y, Medina JSM, Feng Y, Ptashkin RN, Bolton KL, Farnoud N, Zhou Y, Patel MA, Asimomitis G, Cobbs CC, Mohibullah N, Huberman KH, Arcilla ME, Kushner BH, Modak S, Kung AL, Zehir A, Levine RL, Armstrong SA, Cheung NKV, Papaemmanuil E (2022). “Bone marrow surveillance of pediatric cancer survivors identifies clones that predict therapy-related leukemia”. *Clinical Cancer Research* 28, 1614–1627. 10.1158/1078-0432.CCR-21-2451. [\[Paper\]](#)
12. Bolton KL, Koh Y, Foote MB, Im H, Jee J, Sun CH, Safonov A, Ptashkin R, Moon JH, Lee JY, Jung J, Kang CK, Song K-H, Choe PG, Park WB, Kim HB, Oh M-D, Song H, Kim S, Patel M, Derkach A, Gedvilaite E, Tkachuk KA, Wiley BJ, Chan IC, Braunstein LZ, **Gao T**, Papaemmanuil E, Esther NB, Pessin MS, Kamboj M, Diaz LA Jr, Ladanyi M, Rauh MJ, Natarajan P, Machiela MJ, Awadalla P, Joseph V, Offit K, Norton L, Berger MF, Levine RL, Kim ES, Kim NJ, Zehir A (2021). “Clonal hematopoiesis is associated with risk of severe COVID-19”. *Nature Communications* 12, 5975. 10.1038/s41467-021-26138-6. [\[Paper\]](#)
13. Bolton KL, Ptashkin RN*, **Gao T***, Braunstein L, Devlin SM, Kelly D, Patel M, Berthon A, Syed A, Yabe M, Coombs CC, Caltabellotta NM, Walsh M, Offit K, Stadler Z, Mandelker D, Schulman J, Patel A, Philip J, Bernard E, Gundem G, Ossa JEA, Levine M, Martinez JSM, Farnoud N, Glodzik D, Li S, Robson ME, Lee C, Pharoah PDP, Stopsack KH, Spitzer B, Mantha S, Fagin J, Boucai L, Gibson CJ, Ebert BL, Young AL, Druley T, Takahashi K, Gillis N, Ball M, Padron E, Hyman DM, Baselga J, Norton L, Gardos S, Klimek VM, Scher H, Bajorin D, Paraiso E, Benayed R, Arcila ME, Ladanyi M, Solit DB, Berger MF, Tallman M, Garcia-Closas M, Chatterjee N, Diaz LA Jr, Levine RL, Morton LM, Zehir A, Papaemmanuil E (2020). “Cancer therapy shapes the fitness landscape of clonal hematopoiesis”. *Nature Genetics* 52, 1219–1226. 10.1038/s41588-020-00710-0. [\[Paper\]](#)
14. **Gao T***, Gillispie GJ*, Copus JS, Pr AK, Seol YJ, Atala A, Yoo JJ, and Lee SJ (2018). “Optimization of gelatin–alginate composite bioink printability using rheological parameters: a systematic approach”. *Biofabrication* 10, 034106. 10.1088/1758-5090/aacdc7. [\[Paper\]](#)
15. Zhang J, **Gao T**, Maher CA (2017). “INTEGRATE-Vis: a tool for comprehensive gene fusion visualization”. *Scientific Reports* 7. 10.1038/s41598-017-18257-2. [\[Paper\]](#) [\[Code\]](#)

PRESENTATIONS & INVITED TALKS

Invited speaker, Program in Quantitative Genomics (PQG) Working Seminar, Harvard T.H. Chan School of Public Health – Boston, MA	Apr. 2026
Short talk speaker, Keystone Symposia on Hematopoiesis – Keystone, CO	Feb. 2026
Invited speaker, Emerging Leaders in Cancer Research Seminar Series, Roswell Park Cancer Center – Buffalo, NY	Jul. 2024
Speaker, Biomedical Informatics PhD Program Recruitment Event – Boston, MA	Feb. 2024
Speaker, Altos Labs San Diego Institute of Science Annual Retreat – San Diego, CA	Dec. 2023
Poster presenter, Cold Spring Harbor Laboratory Genome Informatics Conference – New York, NY	Dec. 2023
Lightning talk speaker, NIH Somatic Mosaicism across Human Tissues (SMAHT) Network Midyear Meeting	Dec. 2023
Speaker, New York University Applied Bioinformatics Laboratory Single-Cell Club – New York, NY	Sep. 2022
Speaker, Single-Cell Cancer Biology Gordon Research Conference – Easton, MA	Jun. 2022
Speaker, Broad Institute Single-Cell Seminar Series – Cambridge, MA	Feb. 2022
Speaker, American Society of Hematology (ASH) 62nd Annual Meeting and Exhibition – San Diego, CA	Dec. 2020

HONORS & AWARDS

Damon Runyon-HHMI Fellowship Award, Damon Runyon Cancer Research Foundation – New York, NY	Feb. 2025
Rising Young Scientist Award (\$1000), Karin Grunebaum Cancer Research Foundation – Boston, MA	May 2023
1st place, Silicon Valley AI & Genomics Hackathon, Google, NVIDIA and NCBI – San Francisco, CA	Jun. 2017
Thomas Scholarship, Washington University – St. Louis, MO	Sep. 2015

ACADEMIC SERVICE

Mentoring

Krishna Dasari (Undergraduate student, Yale; Summer scholar, Park Lab, Harvard Medical School) Jun. 2023 – Aug. 2023

- Project: Estimating oncogene dosage sensitivity from pairwise comparisons of tumor subclones

Christopher Rouya (Medical Student, Harvard Medical School) Feb. 2025 – Present

- Project: Variant detection in single-cell multiome for lineage tracing

Outreach

Steering Committee Member, Internationals @ Broad affinity group, Broad Institute Feb. 2025 – Present

Scientist in Residence, Broad Discovery Center Meet-a-Scientist Educational Program, Broad Institute Sep. 2024 – Present

Student Committee Member, Bioinformatics and Integrative Genomics Ph.D. program, Harvard University Sep. 2023 – May 2024

Peer Mentor, Bioinformatics and Integrative Genomics Ph.D. program, Harvard University Aug. 2022 – May 2024

Team Lead, Sling Health Incubator Network, Washington University Sep. 2014 – Oct. 2016

Co-founder and Vice President, Washington University Popular Science Magazine, Washington University Sep. 2014 – May 2016

Peer Review

Journals: *Bioinformatics*, *Cell Reports*, *Nature Genetics* (co-reviewer), *Nature Communications*, *Frontiers in Immunology*

Conferences: *BICOB-2022 14th International Conference on Bioinformatics and Computational Biology*

TEACHING EXPERIENCES

Harvard Medical School (Cambridge, MA, USA) Sep. 2022 – Dec. 2022

Teaching Fellow, Concepts in Genome Analysis (BMIF 201)

Washington University (St. Louis, MO, USA) Aug. 2017 – Dec. 2017

Teaching Assistant, Elementary to Intermediate Statistics and Data Analysis (Math 3200)